Lille - M2GL - Compilers
Admin Stuff

Guillermo Polito













The Team (In Alphabetical Order)

- **Romain Degrave:** PhD candidate @ Inria —security and languages
- **Federico Lochbaum:** PhD candidate @ Inria —performance and fuzzing
- **Guille Polito:** Researcher @ Inria — languages, VMs, testing…
- **Pablo Tesone:** Researcher @ Pharo Consortium — languages, VMs…

Looking for an internship/stage ?

- **Contact me:** guillermo.polito@inria.fr

- Topics: testing, compilers, benchmarks, security, isolation…

- We are ***interviewing***

Reality is Complex

- **Compiler Course = Complex Domain + Short Time:**
  - Diving into a **real-world compiler** in a couple of weeks is not feasible
  - Lots of concepts, very technical

- **Unknown public:** you may have, maybe, perhaps, **experience. Or NOT.**

- We all **speak different languages**: french, spanish, *Java, Lisp, Smalltalk*…

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Some Choices

- We will study **fundamental concepts**
- You will explore and learn ***lots*** of languages



- We love **examples**
- **Exploration**-based labs
- Learn by **observing** real world examples
- Not only code, but **reflect on the code** to understand

Rough Agenda 1st part — can/may deviate

- **04/11 - Intro and Stack Machines**
- 11/11 - Holidays — You’re lucky. Not. Finish the exercises!
- 18/11 - Tree -> Stack. Grammars and parsers
- 25/11 - Register Machines. Tree -> Registers
- 02/12 - Function Calls, calling conventions
- 09/12 - Stack Frames
- 16/12 - EXAM - Amphitheater to confirm

Time Table (rough estimate)

- **8h-9h:** Theory — One lecture common for all groups
- **9h-9h15:** Break
- **9h15-12h:** Practice — Each group in a different lab room

Evaluation Formula

$$CC * 0,5 + DS1 * 0,25 + DS2 * 0,25$$

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Strong Focus on Practice and Presence

About Continuous Control

- We prepared **lots** of **practical exercises**
  - Coding, but not only!
  - Also exploring and understanding, explaining, drawing…

- Some will be *optional*, some *mandatory*
- Some will be on *machine*, some on *paper*
- You may need to do some *at home*

Evaluation Formula

CC * 0,5 + DS1 * 0,25 + DS2 * 0,25

Strong Focus on Practice and Presence

(We take these special circumpstances into account)

Evaluation Formula

$$CC * 0{,}5 + DS1 * 0{,}25 + DS2 * 0{,}25$$

Two exams

About Exams

- Will be written on paper
- Exercises similar to those in the practice

  => they will be very easy if you do the practices!

- One at the end of the first period (estimated date — 16/12)
- One at the end of the last period (estimated date — 08/02)

About AI

- We all use AI. You use AI. I (*ai*\) use AI.
- Do we do it right?

- My goal as a person: to become a better person
  - How can I use AI to get better?

- My goal as a teacher: that you learn
  - How can we use AI to learn!?

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AI should not just DO for you.
It should help you get further

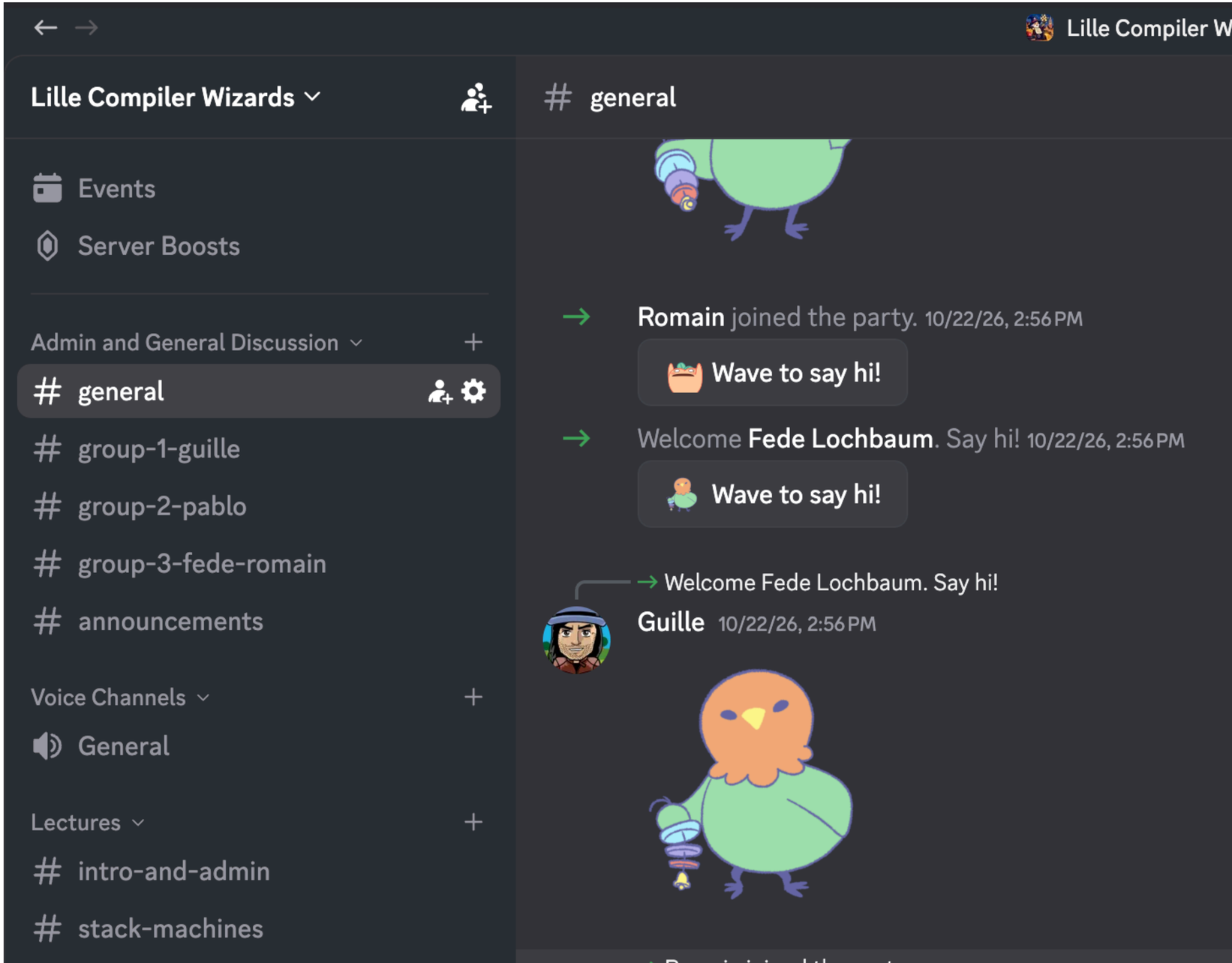
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Discord

- Our communication channel

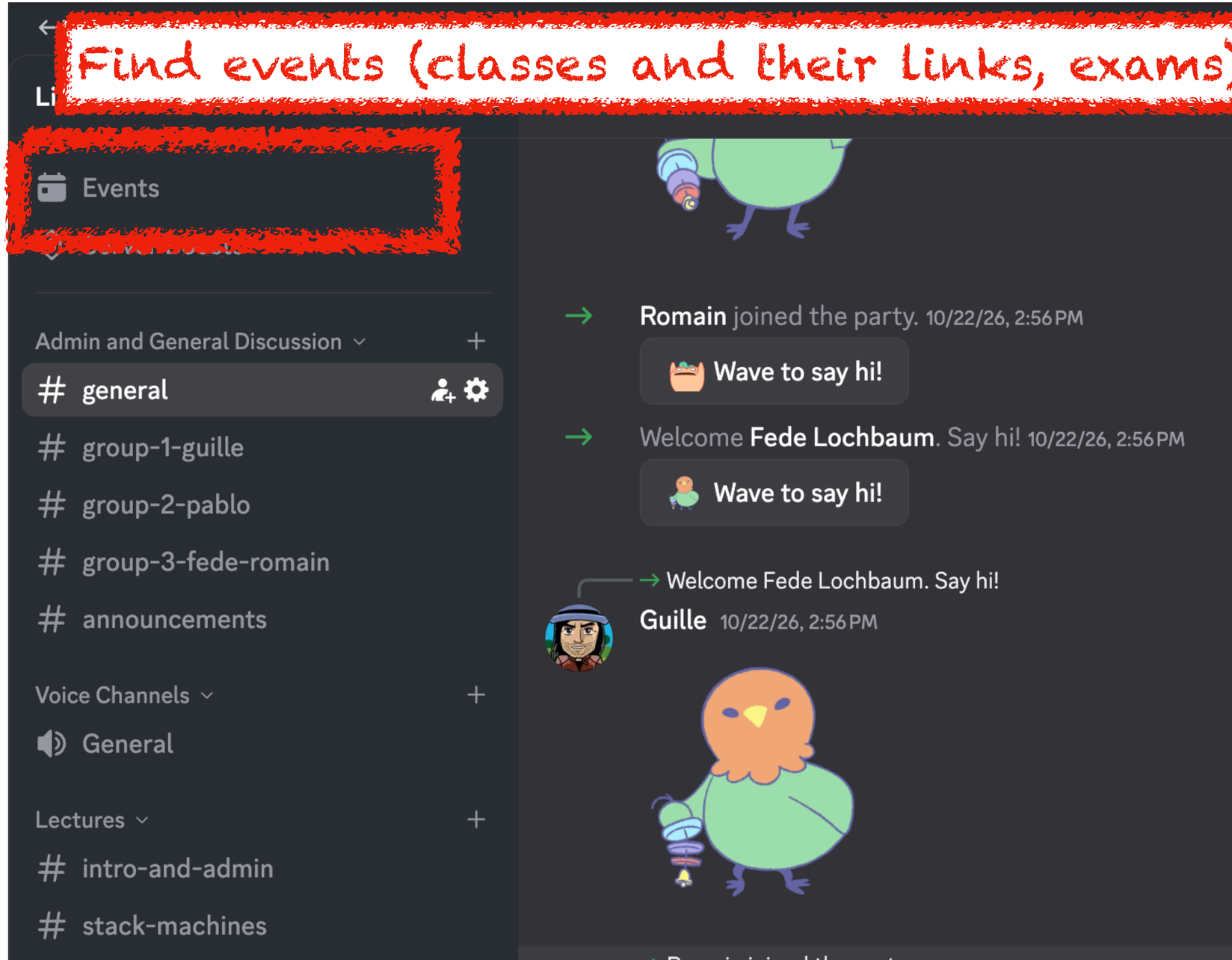
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- Discuss topics
- Ask questions



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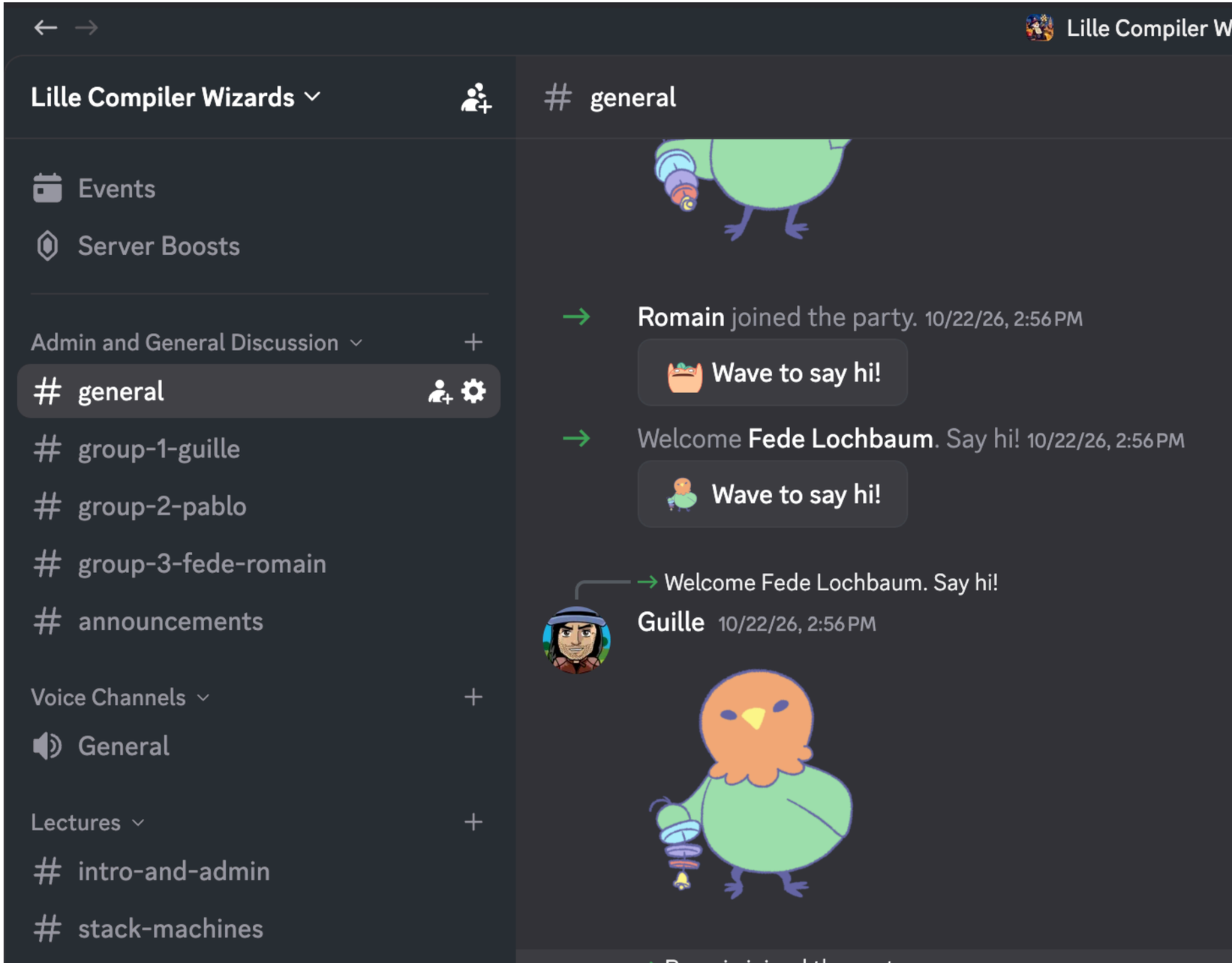
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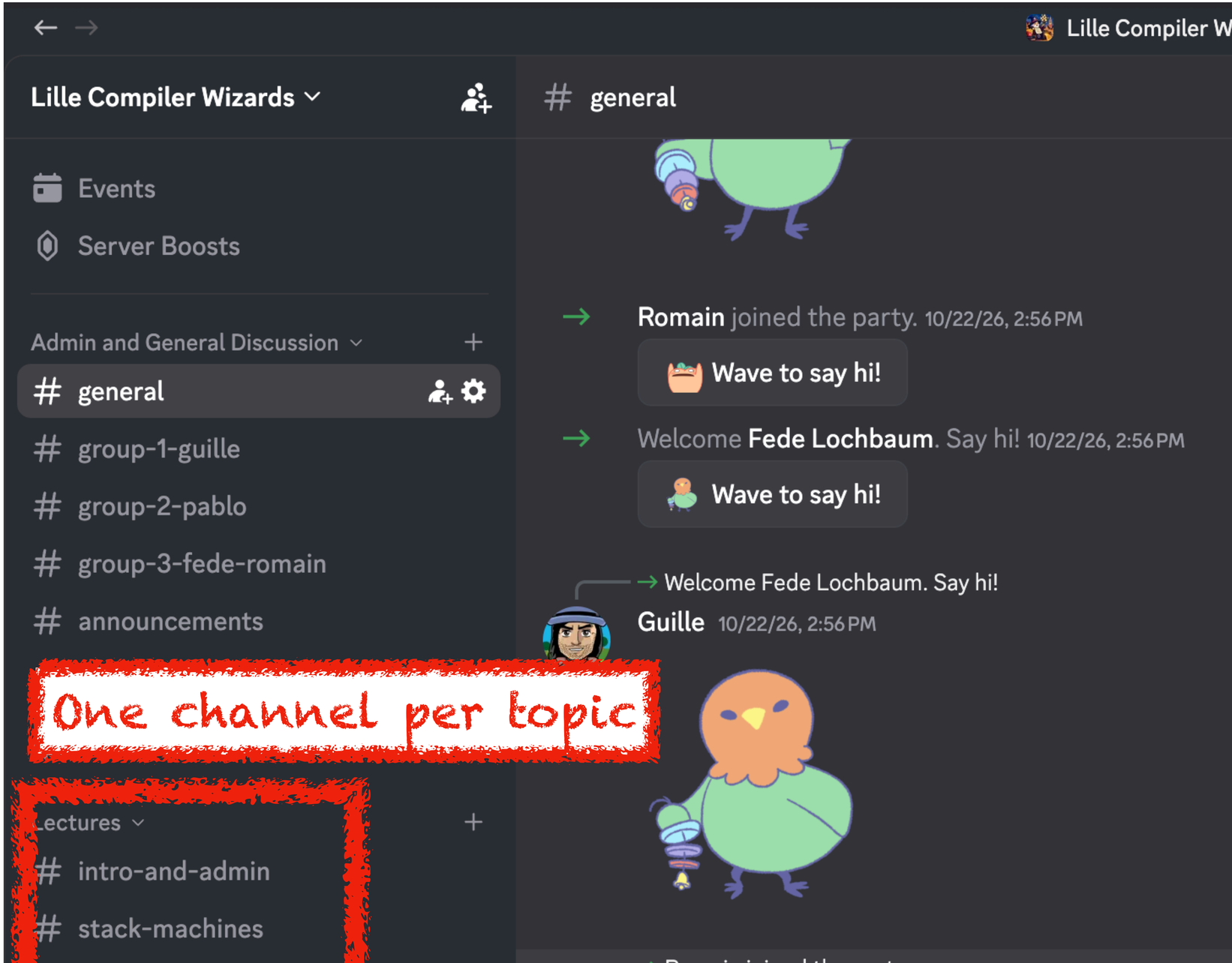
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- https://discord.gg/fMqFeAgTV

